

RANBIR KUMAR

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EDUCATION

B.Tech, Computer Science (AI & ML) — Vellore Institute of Technology, Chennai Aug 2023 – Present
CGPA: 8.28 / 10 | Expected Graduation: May 2027
Relevant Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Explainable AI, Speech & Natural Language Processing, Computer Vision, Machine Vision, Data Structures & Algorithms, Probability & Statistics, SQL

EXPERIENCE

Technical Integration Lead | IORA, Noida May 2026 – Present
• Own technical integration for an nRF9151-based LTE-M/NB-IoT safety wearable; drive architecture reviews across PCB, firmware (MQTT over TLS, GNSS, flash-buffered telemetry), and power subsystems.
• Build IORA AI, a RAG document-intelligence platform: FastAPI + Qdrant vector retrieval with grounded-answer coverage gating, persistent per-user memory, and multi-provider LLM fallback (Gemini, Groq, OpenRouter).

Software Testing & Product Development Intern | IMAI, Delhi May 2026 – Present
Internet and Mobile Association of India
• Execute k6 load, stress, and spike testing on staging web platforms; documented vulnerability findings and consolidated performance reports, improving pre-release reliability.
• Contribute to a GIS-based mapping platform: map feature development, level-of-detail (LOD) visualization, and feature validation for large-scale spatial data.

Computer Vision Engineer (Intern) | Proeffico Solutions Pvt. Ltd., Noida May – Oct 2025
On-site May–Jul 2025; continued remotely Aug–Oct 2025
• Designed and deployed 4 production CV pipelines (speed monitoring, pothole detection, fire detection, workplace safety analytics) on NVIDIA Jetson Xavier NX, achieving <40 ms real-time inference latency, reducing manual monitoring effort and enabling automated real-time decision-making in client workflows.

Team Lead, Rover | Technocrats Robotics, VIT Chennai Aug 2025 – Present
Member since Mar 2024
• Lead the rover autonomy stack: CV-based soil classification and autonomous arrow-marker detection under rough terrain conditions, deployed on Raspberry Pi and an i7 mini-PC.
• Built an end-effector camera pipeline for tool identification and autonomous approach, positioning the arm for manual-trigger pickup.
• All-India Rank 1 (DD Robocon Round 1); International Rover Challenge Finalist — Top 20 (IRC 2025), Top 30 (IRC 2026).

PROJECTS

Semantic Retrieval System & RAG Pipelines (SemantiSearch) MiniLM · GMM · FAISS · FastAPI · Docker
• Built and containerized a semantic retrieval backend as a FastAPI/Docker microservice; validated end-to-end RAG pipeline with OpenAI achieving ~480 ms p95 latency, with confirmed LangChain/LlamaIndex compatibility.
• Implemented GMM-based cluster-partitioned semantic caching to group similar query embeddings, reducing redundant FAISS lookups by ~40% and improving average query response time.

Explainable Customer Intelligence System (CraveConnect) Power BI · XGBoost · SHAP · LIME · Flask
• Built a Power BI-integrated end-to-end ML inference system with SHAP and LIME explainability, enabling interpretable predictions and decision-driven customer segmentation via a deployed Flask API.

Annadata – Agricultural AI Platform (SIH 2025 Finalist) YOLOv8 · ResNet-34 · Flask · Python
• Developed an AI agritech platform using YOLOv8 and ResNet-34, achieving 91% disease classification and 97% soil health prediction accuracy, with GPS-enabled asynchronous pipelines for real-time field monitoring.

RESEARCH & PATENTS

Patrolling Bot — AI-Powered Autonomous Campus Surveillance Robot Patent filed; publication pending
• Invented an autonomous patrolling robot using YOLO-based detection (missing ID cards, fire/smoke, overcrowding) and action-recognition models for fight detection, with real-time alerts to security personnel; embedded deployment on Jetson Xavier NX / Raspberry Pi with LiDAR/ultrasonic navigation (Python, OpenCV, PyTorch, ROS).

Vision-Based Terrain Analysis Accepted; publication forthcoming
• Developed real-time multi-model perception system (YOLOv8 + MiDaS) for terrain classification in autonomous navigation; deployed to embedded edge hardware.

Few-Shot Learning with Explainable AI Peer-reviewed journal, under review
• Designed N-way K-shot classifier on hospital-sourced low-data medical diagnosis; achieved 93% diagnostic accuracy with LIME + SHAP attribution for full prediction interpretability.

SKILLS

AI / ML: RAG Pipeline Design, Semantic Search, Vector Embeddings, Explainable AI (SHAP, LIME), Few-Shot Learning, Reinforcement Learning, Prompt Engineering, LLM Integration (LangChain, LlamaIndex)
Computer Vision: YOLOv8, ResNet, MiDaS, Object Detection, Action Recognition, Depth Estimation, Real-Time Inference, Edge Deployment (NVIDIA Jetson Xavier NX, Raspberry Pi)
ML Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, HuggingFace Transformers, FAISS, Qdrant
Backend & Deployment: FastAPI, Flask, Docker, REST APIs, Microservices, ROS
Data & Tools: Power BI (Dashboarding, DAX, Data Modeling), Pandas, NumPy, OpenCV, SQL, Git, Linux, k6

ACHIEVEMENTS

• Patent filed (Patrolling Bot) | SIH Finalist 2025 | AIR 1, DD Robocon | IRC Finalist — Top 20 (2025), Top 30 (2026)